

Lab: The Time Machine -- Thinking Ahead by Simulating the Future

Objective

Practice **planning** as future-oriented simulation by building decision trees, evaluating counterfactual scenarios, and analyzing how you weigh risk against ambiguity in everyday decisions.

Prerequisites

- Completed all previous Everyday Life modules (Systems through Communication)
- Read the module on Future Simulation

Part 1: Decision Tree Builder

Goal: Map a real upcoming decision as a branching tree of possible futures.

Choose a decision you face in the next two weeks (choosing classes, planning a weekend, deciding whether to try out for a team). Build a decision tree:

1. **Root node:** State the decision.
2. **Branch 1 (Policy A):** What is the first action? What might happen next? Draw at least two levels of consequences.
3. **Branch 2 (Policy B):** Same for the alternative action.
4. **Branch 3 (Policy C):** A third option if one exists.

For each endpoint, rate: - **How much do you want this outcome?** (1-10, where 10 = strongly preferred) - **How uncertain are you that it will happen?** (1-10, where 10 = very uncertain)

Write your response here

Part 2: Counterfactual Thinking

Goal: Practice imagining "what if" scenarios to understand how the brain simulates alternatives.

Think of a decision you made in the past week. Now imagine you had chosen differently:

1. **What you actually did:** Describe the action and the outcome.
2. **What you could have done:** Describe the alternative action.
3. **Imagined outcome:** What do you think would have happened?
4. **Confidence:** How sure are you about the imagined outcome? What are you uncertain about?
5. **Active Inference connection:** When you imagine an alternative future, you are running your generative model forward in time. What does this tell you about the purpose of imagination?

Write your response here

Part 3: Risk vs. Ambiguity

Goal: Distinguish between risk (known probabilities) and ambiguity (unknown probabilities).

Consider two bets:

- **Bet A (Risk):** Flip a fair coin. Heads you win \$20, tails you win nothing. You know the odds are 50/50.

- **Bet B (Ambiguity):** Draw from a mystery bag. You might win \$30 or nothing, but you have no idea what the odds are.
- Which bet would you choose? Why?
- Which bet has higher expected value?
- Which bet has higher ambiguity?
- In Active Inference, why might an agent prefer the risky bet even when the ambiguous bet might pay more?
- Now imagine you can pay \$2 to peek inside the mystery bag first. Would you? This is epistemic action -- paying a cost to reduce ambiguity.

Write your response here

Part 4: Planning Horizon Experiment

Goal: Explore how the depth of future simulation affects decision quality.

Think about a goal you have (getting into college, saving money, learning a skill). Plan it at three different horizons:

Planning Horizon	Plan	How Detailed?	How Confident?
Next week			
Next month			
Next year			

1. Which horizon produced the most detailed plan?
2. Which horizon had the most uncertainty?
3. Why does the brain struggle to plan far into the future?

Write your response here

Summary Table

Concept	Definition	Your Example
Planning	Simulating future action sequences to select the best policy	
Counterfactual Thinking	Imagining alternative outcomes that did not actually happen	
Risk	Uncertainty with known probabilities	
Ambiguity	Uncertainty with unknown probabilities	
Planning Horizon	How far into the future the agent simulates	